

Topic B2

Cells

Concept guide · Worked application · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, biological mechanisms, common misconceptions, data interpretation and original exam-style practice for this topic.

Syllabus map

B2.1	Cell structure and organisation
B2.2	Levels of organisation
B2.3	Size of specimens

How to study: retrieve definitions, cover the application, then complete the practice paper without using the notes.

1. Cell structures

The cell membrane controls movement of substances; cytoplasm is the site of many reactions; the nucleus contains genetic material; mitochondria carry out aerobic respiration; ribosomes synthesise proteins. Plant cells additionally have a cellulose wall, permanent vacuole and chloroplasts in photosynthetic tissues.

Examiner insight

Do not state that every plant cell has chloroplasts; root cells normally do not.

2. Specialised cells

Structure is adapted to function: root hair cells have a long extension for absorption; red blood cells are biconcave and lack a nucleus; sperm cells have a flagellum and many mitochondria; palisade cells contain many chloroplasts; ciliated cells move mucus.

Examiner insight

Name a feature, then explain exactly how it improves the function.

3. Levels of organisation

Cells of similar function form tissues; tissues combine into organs; organs cooperate in organ systems; systems make an organism. For example, muscle and glandular tissues form part of the stomach, which functions in the digestive system.

Examiner insight

An organ contains several tissue types, not just many identical cells.

4. Magnification

Magnification compares image size with actual size. Use consistent units before dividing. One millimetre equals 1000 micrometres.

Examiner insight

magnification = image size / actual size. Rearrange before substituting and include units in the converted quantities.

Worked application

Situation: A micrograph shows a cell image 40 mm long at magnification x800.

Prediction: Actual length is 0.050 mm or 50 micrometres.

Reason: Actual size equals image size divided by magnification after unit conversion.

Exam technique

High-value habits

Define terms precisely, describe visible patterns before explaining them, link structure to function, use data values as evidence, identify variables explicitly and answer the command word directly.

Retrieval checklist

- Recall five key definitions without notes.
- Explain one process as a linked cause-and-effect chain.
- Interpret the data set and identify a limitation.
- Complete the timed practice and mark only after finishing.

Practice Paper

Topic B2

rganisation of the organism

Instructions

Answer all questions. Select one option for multiple choice. Use values from tables where relevant. Write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A - Multiple choice

Choose the best answer for each question.

1. Which statement best defines tissue? [1]
 - A a group of cells with similar structures working together for a function
 - B a structure made of different tissues working together
 - C a specialised structure within a cell
 - D image size divided by actual size
2. Which statement best defines organ? [1]
 - A a specialised structure within a cell
 - B a structure made of different tissues working together
 - C image size divided by actual size
 - D the ability to distinguish two close points as separate
3. Which statement best defines organelle? [1]
 - A image size divided by actual size
 - B the ability to distinguish two close points as separate
 - C a specialised structure within a cell
 - D a group of cells with similar structures working together for a function
4. Which statement best defines magnification? [1]
 - A the ability to distinguish two close points as separate
 - B a group of cells with similar structures working together for a function
 - C a structure made of different tissues working together
 - D image size divided by actual size
5. Which statement best defines resolution? [1]
 - A the ability to distinguish two close points as separate
 - B a group of cells with similar structures working together for a function
 - C a structure made of different tissues working together
 - D a specialised structure within a cell
6. Which statement about cell structures is correct? [1]
 - A A group of organs.
 - B The cell membrane controls movement of substances; cytoplasm is the site of many reactions; the nucleus contains genetic material; mitochondria carry out aerobic respiration; ribosomes synthesise proteins.
 - C One specialised cell.
 - D A collection of organ systems.

7. Which statement about specialised cells is correct? [1]
- A One specialised cell.
 - B A collection of organ systems.
 - C Structure is adapted to function: root hair cells have a long extension for absorption; red blood cells are biconcave and lack a nucleus; sperm cells have a flagellum and many mitochondria; palisade cells contain many chloroplasts; ciliated cells move mucus.
 - D Actual size divided by image size.
8. Which statement about levels of organisation is correct? [1]
- A A collection of organ systems.
 - B Actual size divided by image size.
 - C How large an image appears.
 - D Cells of similar function form tissues; tissues combine into organs; organs cooperate in organ systems; systems make an organism.
9. Which statement about magnification is correct? [1]
- A Magnification compares image size with actual size.
 - B Actual size divided by image size.
 - C How large an image appears.
 - D A group of organs.
10. Which prediction is most appropriate when a micrograph shows a cell image 40 mm long at magnification x800? [1]
- A A group of organs.
 - B A collection of organ systems.
 - C Actual length is 0.050 mm or 50 micrometres.
 - D How large an image appears.

Section B · Structured questions

(a) Define tissue.

[2]

(b) Define organ.

[2]

(c) Distinguish organelle from the incorrect idea: “a collection of organ systems”.

[2]

(d) State two important features of cell structures.

[2]

[Total: 8]

12 A micrograph shows a cell image 40 mm long at magnification x800.

(a) State the most likely biological outcome.

[2]

(b) Explain the outcome using a linked biological mechanism.

[3]

(c) Predict what would happen if the relevant change were reversed. [2]

(d) State one assumption made in this prediction. [1]

[Total: 8]

13 A student records image length under three conditions.

condition	image length
specimen X	18
specimen Y	32
specimen Z	46

(a) Calculate the difference between the highest and lowest recorded values. [2]

(b) Describe the pattern in the results. [2]

(c) Explain the pattern using biological knowledge. [3]

(d) State one limitation of drawing a conclusion from these data. [1]

[Total: 8]

14 A student plans to prepare and observe onion epidermal cells using a light microscope.

(a) Identify the independent variable. [1]

(b) Identify the dependent variable. [1]

(c) State two control variables. [2]

(d) Describe how to obtain reliable results across a suitable range. [3]

(e) Suggest one specific improvement. [1]

[Total: 8]

15 Apply your knowledge of organisation of the organism to an unfamiliar context.

(a) Define magnification. [2]

(b) Explain the biological importance of levels of organisation. [3]

(c) Correct this statement: “how large an image appears”.

[2]

(d) State one further question that could be investigated.

[1]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award points for accurate biology, use of evidence and linked reasoning, then diagnose the first error.

Solutions · Multiple choice

- 1 A. A group of cells with similar structures working together for a function.
- 2 B. A structure made of different tissues working together.
- 3 C. A specialised structure within a cell.
- 4 D. Image size divided by actual size.
- 5 A. The ability to distinguish two close points as separate.
- 6 B. Do not state that every plant cell has chloroplasts; root cells normally do not.
- 7 C. Name a feature, then explain exactly how it improves the function.
- 8 D. An organ contains several tissue types, not just many identical cells.
- 9 A. $\text{magnification} = \text{image size} / \text{actual size}$. Rearrange before substituting and include units in the converted quantities.
- 10 C. Actual size equals image size divided by magnification after unit conversion.

Solutions · Structured questions

11. (a) a group of cells with similar structures working together for a function [2]. (b) a structure made of different tissues working together [2]. (c) a specialised structure within a cell; not a collection of organ systems [2]. (d) Credit any two accurate features from the guide: The cell membrane controls movement of substances; cytoplasm is the site of many reactions; the nucleus contains genetic material; mitochondria carry out aerobic respiration; ribosomes synthesise proteins. Plant cells additionally have a cellulose wall, permanent vacuole and chloroplasts in photosynthetic tissues. [2].

12. (a) Actual length is 0.050 mm or 50 micrometres [2]. (b) Actual size equals image size divided by magnification after unit conversion [3]. (c) Accept the biologically justified opposite trend when the causal factor is reversed [2]. (d) One valid assumption, such as other relevant conditions remaining constant [1].

13. (a) $46 - 18 = 28$ [2]. (b) The values change from 18 to 46, with the highest value 46. [2]. (c) Larger images do not necessarily mean larger actual cells unless magnification is the same [3]. (d) Only three conditions / no repeats / small dataset / uncontrolled variation [1].

14. (a) Stain used or tissue location [1]. (b) Number of clearly visible cells [1]. (c) Same magnification, sample area and light level [2]. (d) Use several values across a sensible range, repeat each condition, identify anomalies and calculate means [3]. (e) Lower the coverslip at an angle to avoid air bubbles [1].

15. (a) image size divided by actual size [2]. (b) Award linked points from: Cells of similar function form tissues; tissues combine into organs; organs cooperate in organ systems; systems make an organism. For example, muscle and glandular tissues form part of the stomach, which functions in the digestive system. [3]. (c) Correct statement: resolution means the ability to distinguish two close points as separate [2]. (d) Any testable relevant biological question [1].

Self-assessment

Classify the first lost mark as definition, biological knowledge, causal link, data use, experimental design or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.